

Question 1

A platinum resistance sensor is used to interpolate between the triple point of water (0 °C), the boiling point of water (100 °C) and the freezing point of zinc (419.6 °C). The corresponding resistance values are 100.0 Ω , 138.5 Ω and 253.7 Ω . The algebraic form of the interpolation equation is:

$$R_T = R_0(1 + \alpha T + \beta T^2)$$

where R_T Ω = resistance at T °C
 R_0 Ω = resistance at 0 °C
 α, β = constants.

Find the numerical form of the interpolation equation.

Solution

Point 1 (Triple point of water): $T_1 = 0, R_{T1} = 100$

Point 2 (Boiling point of water): $T_2 = 100, R_{T2} = 138.5$

Point 3 (Freezing point of Zinc): $T_3 = 419.6, R_{T3} = 253.7$

Solve for Constant R_0

Substitute the data from **Point 1** into the general equation:

$$100 = R_0(1 + \alpha \cdot 0 + \beta \cdot 0^2)$$

It can be directly concluded that:

$$R_0 = 100\Omega$$

Substitute Point 2 (100 °C, 138.5 Ω):

$$138.5 = 100(1 + \alpha \cdot 100 + \beta \cdot 100^2)$$

$$\alpha \cdot 100 + \beta \cdot 100^2 = 0.385$$

Substitute Point 3 (419.6 °C, 253.7 Ω):

$$253.7 = 100(1 + \alpha \cdot 419.6 + \beta \cdot 419.6^2)$$

$$\alpha \cdot 419.6 + \beta \cdot 419.6^2 = 1.537$$

$$\beta = -5.8507 \times 10^{-7}, \alpha = 3.9085 \times 10^{-3}$$

$$R_T = 100(1 + 3.9085 \times 10^{-3}T - 5.8507 \times 10^{-7}T^2)$$

Question 2

The following results were obtained when a pressure transducer was tested in a laboratory under the following conditions:

- I Ambient temperature 20 °C, supply voltage 10 V (standard)
- II Ambient temperature 20 °C, supply voltage 12 V
- III Ambient temperature 25 °C, supply voltage 10 V

Input (barg)	0	2	4	6	8	10
Output (mA)						
I	4	7.2	10.4	13.6	16.8	20
II	4	8.4	12.8	17.2	21.6	28
III	6	9.2	12.4	15.6	18.8	22

- (a) Determine the values of K_M , K_I , a and K associated with the generalised model equation $O = (K + K_M I_M)I + a + K_I I_I$.
- (b) Predict an output value when the input is 5 barg, $V_S = 12$ V and ambient temperature is 25 °C.

Solution

(a)

Step 1: Analyze Condition I (Standard State) to find K and a

Condition I (20 °C, 10 V) is defined as the standard operating environment. In this state, there are no environmental changes, so $I_M = 0$ and $I_I = 0$. The model simplifies to an ideal straight line: $O = K \cdot I + a$.

Using data from "Condition I" in the table:

- When input $I = 0$, output $O = 4$ mA. Substituting into the equation: $4 = K \times 0 + a \rightarrow a = 4$ mA.
- When input $I = 10$, output $O = 20$ mA. Using the known value of a :

$$20 = K \times 10 + 4$$

$$10K = 16 \rightarrow K = 1.6 \text{ mA/barg}$$

Step 2: Analyze Condition II (Voltage Change) to find K_M

In Condition II (20 °C, 12 V), the temperature is standard ($I_I = 0$), but the supply voltage has changed.

- **Modifying input change:** $I_M = \Delta V_S = 12 \text{ V} - 10 \text{ V} = 2 \text{ V}$.

- **Simplified Model:** $O = (K + K_M \cdot I_M)I + a$.

Using data from "Condition II" in the table:

- When $I = 0$, $O = 4$ mA. This confirms a is unchanged and voltage does not act as an interfering input.
- When $I = 10$, $O = 28$ mA. Substituting into the equation:

$$28 = (1.6 + K_M \times 2) \times 10 + 4$$

$$24 = (1.6 + 2K_M) \times 10$$

$$2.4 = 1.6 + 2K_M \rightarrow 0.8 = 2K_M \rightarrow \mathbf{K_M = 0.4 \text{ mA/(barg} \cdot \text{V)}}$$

Step 3: Analyze Condition III (Temperature Change) to find K_I

In Condition III (25 °C, 10 V), the voltage is standard ($I_M = 0$), but the temperature has changed.

- **Interfering input change:** $I_I = \Delta T = 25 \text{ °C} - 20 \text{ °C} = \mathbf{5 \text{ °C}}$.
- **Simplified Model:** $O = K \cdot I + a + K_I \cdot I_I$.

Using data from "Condition III" in the table:

- When $I = 0$, $O = 6$ mA. The zero point has shifted, confirming temperature acts as an interfering input.

$$6 = 1.6 \times 0 + 4 + K_I \times 5$$

$$2 = 5K_I \rightarrow \mathbf{K_I = 0.4 \text{ mA/°C}}$$

(b)

Requirement: Predict the output current O for an input pressure $I = 5$ barg, supply voltage $V_S = 12$ V, and environmental temperature $T = 25 \text{ °C}$.

1. **Calculate environmental changes:**

- $I_M = 12 \text{ V} - 10 \text{ V} = 2 \text{ V}$
- $I_I = 25 \text{ °C} - 20 \text{ °C} = 5 \text{ °C}$

2. **Substitute all constants into the full model:**

$$O = (K + K_M \cdot I_M)I + a + K_I \cdot I_I$$

$$O = (1.6 + 0.4 \times 2) \times 5 + 4 + (0.4 \times 5)$$

$$O = (1.6 + 0.8) \times 5 + 4 + 2$$

$$O = 2.4 \times 5 + 6$$

$$O = 12 + 6 = \mathbf{18 \text{ mA}}$$